



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2017/2018

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE IN ELECTRICAL AND ELECTRONIC ENGINEERING
AND BACHELOR OF SCIENCE IN BIOMEDICAL ENGINEERING

EEE 212: MECHANICS OF MACHINES

DATE: Wednesday, 28th February 2018

TIME: 11.00 a.m. - 1.00 p.m.

INSTRUCTIONS:

- 1) This paper contains FIVE questions.
- 2) QUESTION ONE is COMPULSORY and carries 30 marks.
- 3) Answer QUESTION ONE and any other two questions.
- 4) All symbols have usual meaning unless otherwise stated.

QUESTION ONE [COMPULSORY]-30 MARKS

- (a) Define the following terms as used in strength of materials [3 marks]
 - (i) Stress
 - (ii) Strain
 - (iii) Ductility
- (b) Using a well labelled diagram, differentiate between tangent Modulus and secant modulus [4 marks]
- (c) Using a well labeled diagram, illustrate three types of loads as found in strength of materials [6 marks]
- (d) Determine the reactions at support A and B for the overhanging beam subjected to the loading as shown in Fig. Q1 (d). [4 marks]

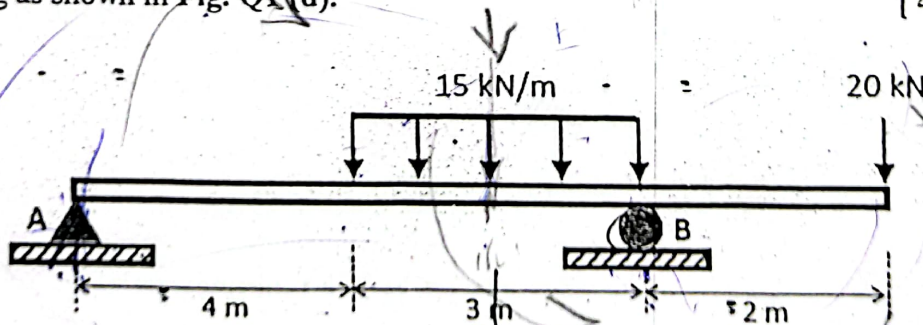


Fig. Q1 (d).

(e) List four factors that affect the coefficient of friction between the belt and the pulley. [2 marks]

(f) Define the term "chain drives" giving four advantages and two disadvantages of chain drives. [4 mark]

(g) A mass of 75 kg is suspended from a spring of stiffness 15kN/m of extension. When the mass is slightly displaced from equilibrium, it vibrates freely with an amplitude of 36mm. Determine;

- (i) The periodic time of the system $t = \frac{2\pi}{\sqrt{g/m}}$ [2 mark]
 (ii) The frequency of vibration $\frac{1}{t}$ [1 mark]
 (iii) The velocity and acceleration when the mass is displaced 16 mm from the equilibrium position. [2 marks]
 (iv) Maximum velocity and acceleration attained. [2 marks]

QUESTION TWO [20 MARKS]

(a) Show that the volumetric strain in a material under axial loading is given by

$$\epsilon_v = \epsilon_x + \epsilon_y + \epsilon_z$$

[5 marks]

(b) For the beam shown in Fig. Q 2(b) below;

[15 marks]

- (i) Determine the reaction forces at the supports ✓
 (ii) Draw the shearing force and bending moment diagrams ✓
 (iii) Calculate the position and magnitude of the maximum bending moment.

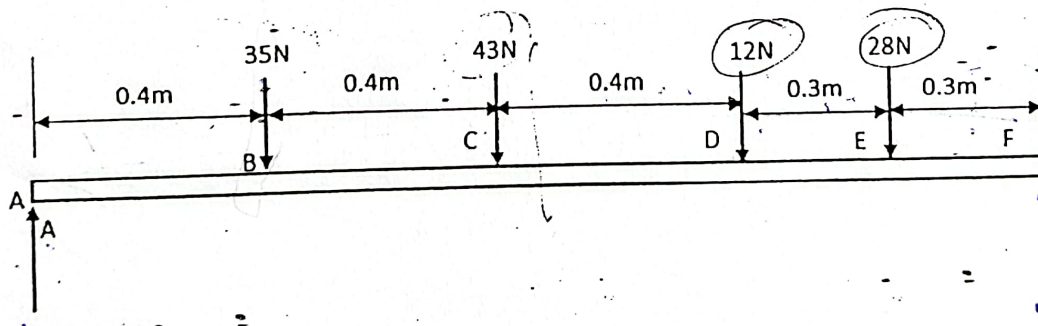


Fig. Q 2(b)

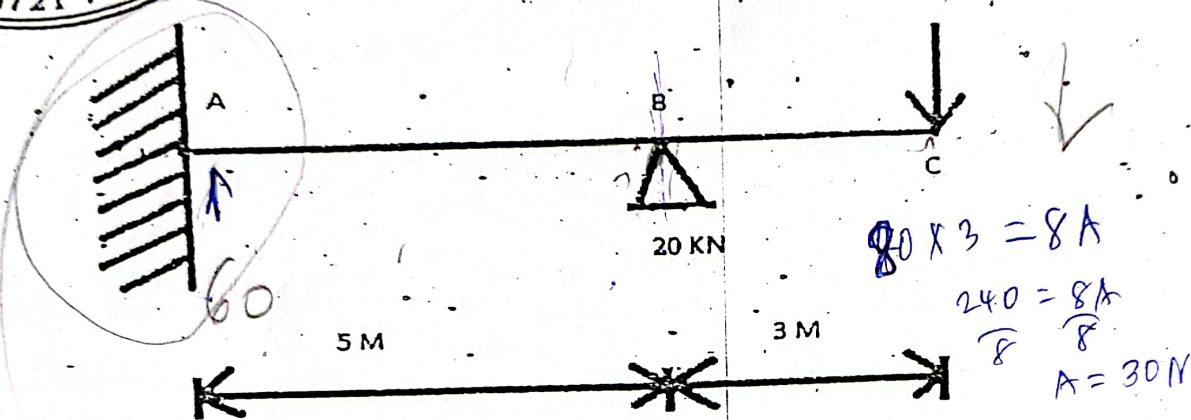
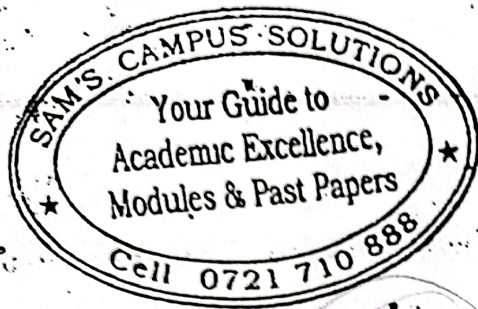
a) State Hooke's Law

EEF 212 / 14/12/2013 (2 Marks)

b) The figure below show a beam. i) Find Reaction A.

ii) Draw S.F.D and B.M.D.

iii) What is the Maximum bending moment?

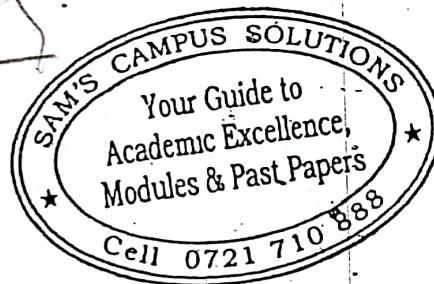


(18 Marks)

Question 4

a) Define the following terms:

- Period
- Frequency
- Free Vibration
- Forced Vibration



(4 Marks)

b) A spring of stiffness $4 \frac{\text{KN}}{\text{m}}$ is suspended vertically and two equal masses of 6 Kg each are attached to the lower end. One of these masses is suddenly removed and the system Oscillate. Determine

- Amplitude and frequency of vibration
- The Velocity and acceleration of mass when passing through half amplitude position
- Energy of Vibration in Joules

(16 Marks)

Question 5

a) Name at least any three moving component of a mechanism?

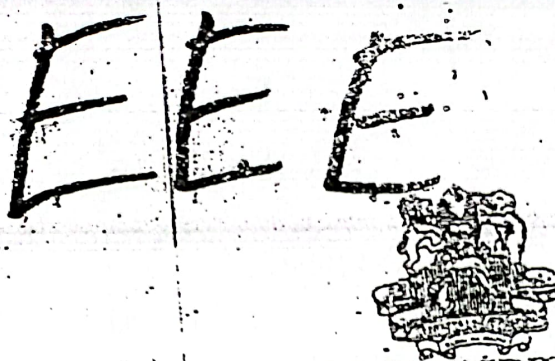
(3 Marks)

b) Discuss any three means through which power can be transmitted?

(3 Marks)

c) What is the angular acceleration, α of 60 kg(cylindrical) pulley of radius $R=0.2 \text{ m}$ and the tension in the cable. If 30 kg block is attached to the end of the cable?

(14 Marks)



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2016/2017

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE IN ELECTRICAL AND ELECTRONIC ENGINEERING AND BACHELOR
OF SCIENCE IN BIOMEDICAL ENGINEERING

EEE 212: MECHANICS OF MACHINES

DATE: Thursday 15th December 2016

TIME: 11.00am-1.00pm

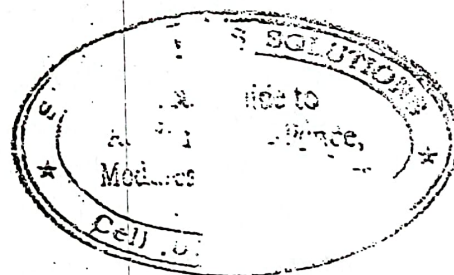
INSTRUCTIONS:

- This paper contains FIVE questions.
- Question 1 is compulsory and then respond to TWO other questions.

Question 1 (30 Marks)

a) Define the following terms: (8 Marks (1 each))

- Mechanics
- Machine
- Rigid body
- Particle
- Statics
- Dynamics
- Kinetics
- Kinematics



b) Define a force and state the characteristics required to completely define the force. (5 Marks)

c) A steel rod of 3 cm diameter is enclosed centrally in a hollow copper tube of external diameter 5 cm and internal diameter of 4 cm. The composite bar is then subjected to an axial pull of 45kN. If the length of each bar is

equal to 15 cm, determine: (E of steel = $2.1 \times 10^5 \text{ N/mm}^2$ and for copper = $1.1 \times 10^5 \text{ N/mm}^2$)

- i. Stresses in the rod and the tube
- ii. Load carried by each bar (7-Marks)

d) A vehicle to be used for establishing a space workshop consists of two parts A and B that are to be separated during flight. At the instant prior to separation, the combined vehicle has a velocity

$$\vec{v} = 5000\hat{i} + 4000\hat{j} + 3000\hat{k} \text{ m/s}$$

At time t after separation, measurements taken on the separated portion A indicated its acceleration components in the three rectangular coordinate directions to be; $a_x = 4.8t \text{ m/s}^2$, $a_y = 1.2t^2 \text{ m/s}^2$, $a_z = 2t \text{ m/s}^2$. It is assumed that the velocity of the main part B of the vehicle remains reasonably constant throughout. Determine:

- i. The velocity of the separated portion A; 20 seconds after separation.
- ii. The distance between the two parts after this time has elapsed. (10 Marks)

$$m \cdot a = F \times A$$

Question 2 (20 Marks)

With the aid sketch of the stress strain graph, explain the behaviour of a mild steel rod in an extensometer. (10 Marks)

Define shear force and bending moment. (4 Marks)

Define 'factor of safety' and explain its importance. (6 Marks)

Question 3 (20 Marks)

- a) i) State Newton's second law of motion. (1 Mark)
- ii) State D'Alembert's principle. (1 Mark)

b) Two blocks of weight W_1 and W_2 are connected by inextensible wire passing over a smooth pulley as shown in Figure 3(a). If W_1 is greater

than W_2 , find the tension in the string and determine the acceleration of the system. (6 Marks)

c) The two crates in Fig. 3(b) are released from rest. Their masses are $M_A = 40\text{kg}$ and $M_B = 30\text{kg}$. If the static and kinetic coefficients of friction between crate A and the inclined surface are $\mu_s = 0.2$ and $\mu_k = 0.15$ respectively, determine:

- Whether the crates will start moving.
- The acceleration of the crates if they are in motion. (12 Marks)

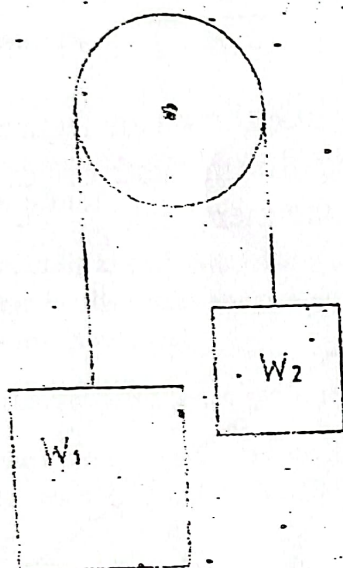


Figure 3 (a)

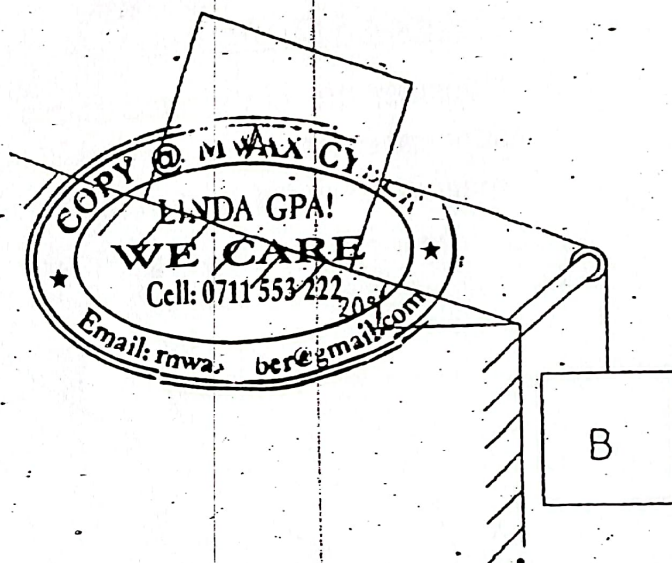


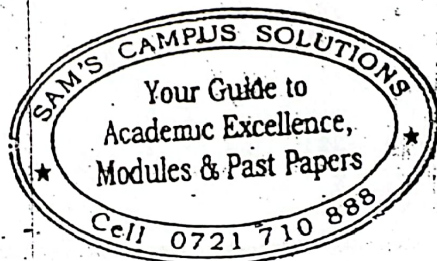
Figure 3 (b)

Question 4 (20 Marks)

a) Figure 4(a) shows the plane area of a sheet metal plate with a circular hole. Determine the coordinates $\bar{x}$ and $\bar{y}$ of the centre of the hole so that this point will be the centroid of the remaining shaded area. (9 Marks)

b) Determine the second moment of area of the cross-section shown in figure 4(b):

- about the horizontal centroid axis.
- about the vertical centroid axis. (11 Marks)



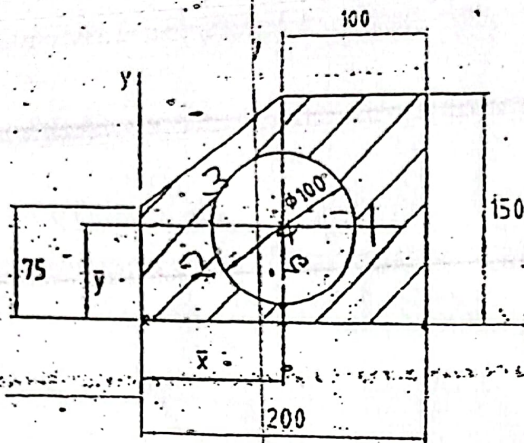


Figure 4 (a)

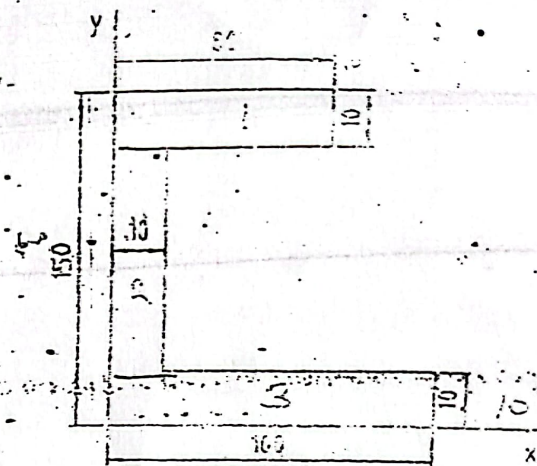


Figure 4 (b)

Question 5 (20 Marks)

Consider the pin-jointed frame ABCDEFG shown in figure 5. Determine the support reactions at A and E and using the method of joints, determine the magnitude and nature of the forces in all the members.

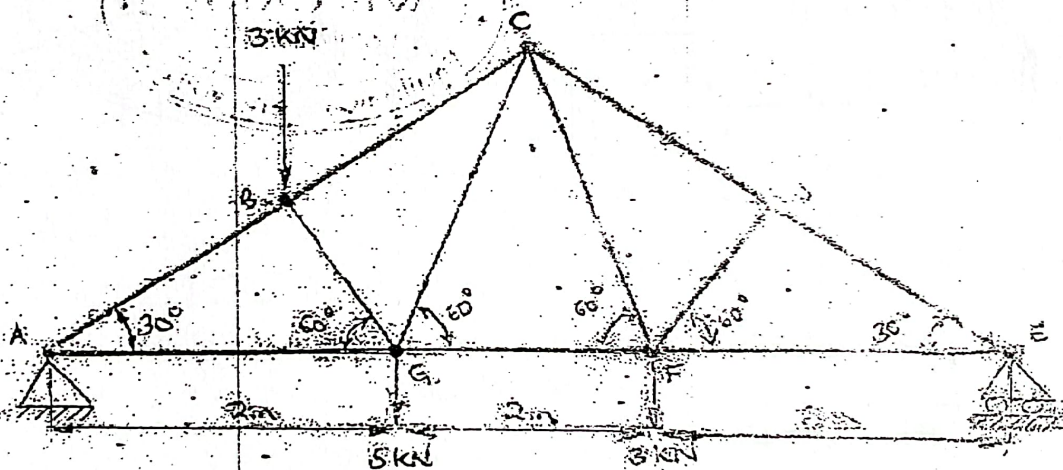
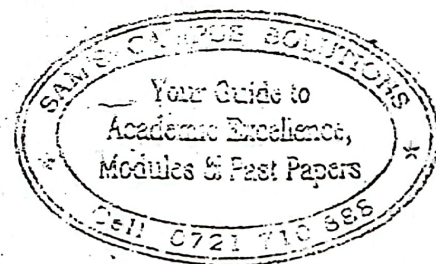


Figure 5



Date: October 2016

Time: 2 hours

1. Define the following terms: (6 Marks)

- a) Machine
- b) Particle
- c) Rigid body
- d) Kinetics
- e) Kinematics
- f) Statics

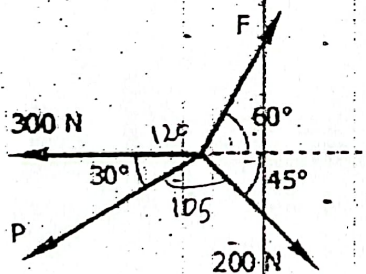


2. Determine the magnitude of R and F in Fig. 1 (a) necessary to keep the force system in equilibrium. (4 Marks)

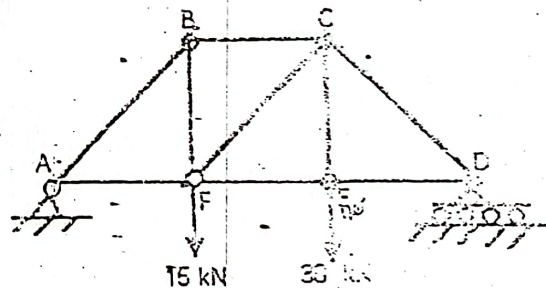
3. The diagram in Fig 1 (b) shows a plane pin-jointed framework subjected to vertical loads of 15kN and 30kN. Solve for the forces and reactions in the frame. The lengths of the vertical and horizontal members are 3m. (14 Marks)

4. Given the loading of the beam as shown in Fig 1 (c), determine the support reactions. (5 Marks)

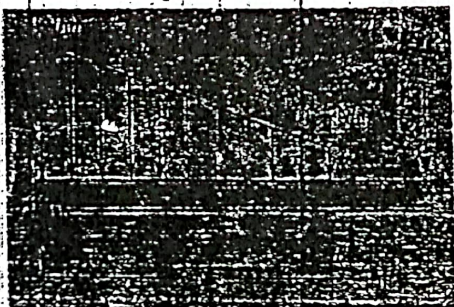
5. Calculate the values and draw the diagrams for shear force and bending moment for the overhanging beam shown in Fig 1 (d). (10 Marks)



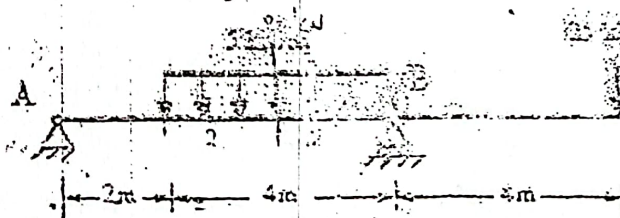
(a) Question 2



(b) Question 3

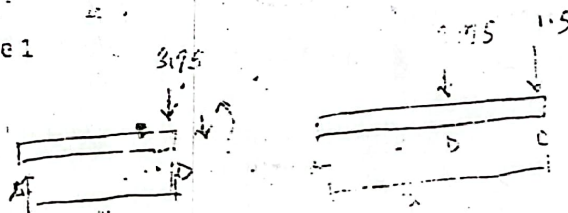


(c) Question 4



(d) Question 5

Figure 1



$$\sum F_y = 15 + 30 + D_y = 0$$

$$D_y = -45$$

$$\sum M_A = 0$$

$$A_x \times 2 = 0$$

$$-3.75 \times 4 + 1.5 \times 8 = 0$$

$$A_x = 0$$



KENYATTA UNIVERSITY
Continuous Assessment Test 1
EEE 212: MECHANICS OF MACHINES

DATE: 13th December 2019

TIME: 1 HOUR

INSTRUCTIONS: ATTEMPT ALL QUESTIONS

- Q1 (i) Describe how a tensile test is carried out (4 Marks)
(ii) Using a stress-strain curve, describe any two properties of engineering materials (4 Marks)

- Q2 The simply supported beam shown in Figure 1 carries a uniformly distributed load of 10 kN/m over entire length and a concentrated load of 80 kN at B.

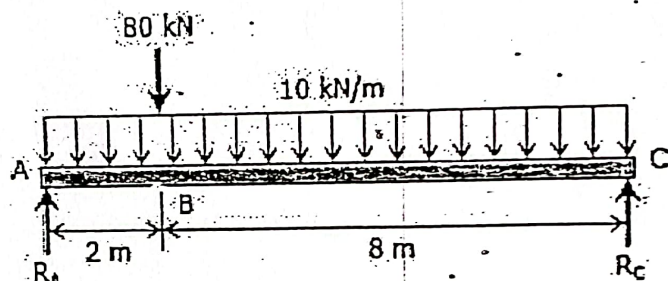


Figure 1

- (i) Compute the reactions at the supports. (2 Marks)
(ii) Derive the expressions for the shear force and the bending moment for each segment of the beam. (4 Marks)
(iii) Sketch the shear force and bending moment diagrams on the graph paper provided, to a linear scale of 1 cm = 1 m and a force scale of 1 cm = 10 kN. Neglect the weight of the beam. (4 Marks)
- Q3 A steel of rectangular cross-section 10 cm deep and 5 cm wide is bent in the planes of the longer side. Estimate the greatest allowable bending moment if the bending stress are not to exceed 150 MN/m² in tension and compression (6 Marks)
- Q4 A steel shaft 35 mm in diameter and 1.2 m long held rigidly at one end has a hand wheel 500 mm in diameter keyed to the other end. The modulus of rigidity of steel is 80 GPa.
- What load applied to tangent to the rim of the wheel produce a torsional shear of 60 MPa? (4 Marks)
 - How many degrees will the wheel turn when this load is applied? (3 Marks)

KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2013/2014

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
(ELECTRICAL AND ELECTRONICS ENGINEERING)

EEE 212: MECHANICAL ENGINEERING

DATE: WEDNESDAY 4TH DECEMBER 2013 Your Guide to TIME 11:00 A.M. - 1:00 P.M.

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Cell 0721 710 888

INSTRUCTIONS:

Answer QUESTION 1 and ANY OTHER TWO questions. QUESTION 1 carries 30 marks while the OTHERS carry 20 marks each.

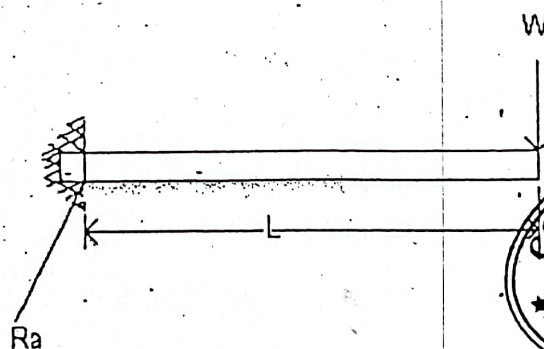
Question 1

a) Use appropriate sketches to explain the following terms

- Stress
- Strain
- Young modulus of elasticity
- Shear Force diagram

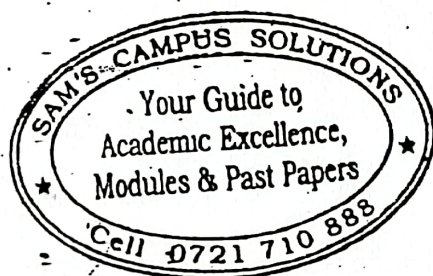
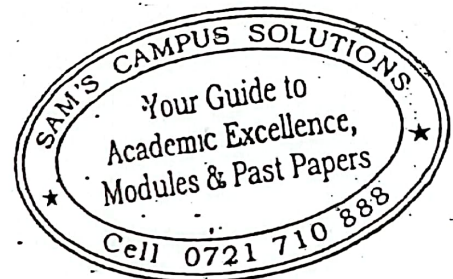
b) Name and explain four types of mechanical loading ?

c) The figure below shows a Uniform cantilever of length (L) and has concentrated load (W) at the free end. Determine



- Shear force at Reaction (R_a) and the bending moment
- Sketch shear force and bending moment diagram

(7 Marks)



What is a mechanism?

(2 Marks)

e) Name atleast two application of shear stress

(2 marks)

f) An Aluminum alloy tube is of 30mm diameter is wound on a drum of 3m diameter. Calculate the maximum bending stress in the tube. Take $E = 80 \frac{GN}{m^2}$

(3 Marks)

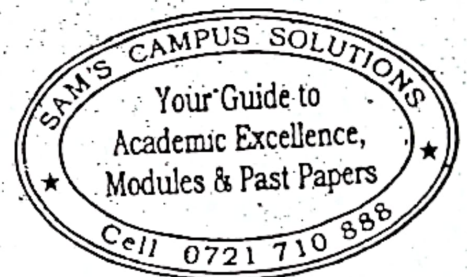
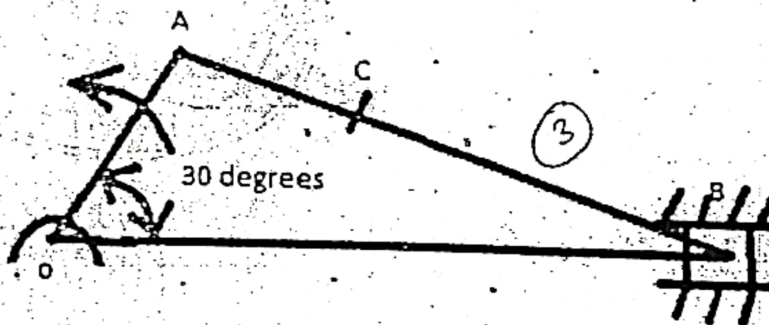
g) A crank OA of the engine mechanism shown in figure below rotates at 3000 rev/min anticlockwise. OA=100mm, and the connecting rod is AB is 200mm long. Find

a) The Piston velocity

b) Angular Velocity of AB

c) The velocity of point C on the rod 50mm from A

(6 Marks)



Question 2

a) Explain the following terms

i) Bending moment

ii) Thermal strain

iii) Bending moment diagram

(6 Marks)

c) State five assumption made in pure bending stresses

(5 Marks)

b) A light-alloy beam, rectangular section 5mm depth, 12 mm wide rest on a support 200mm apart and carries a central load 70 N. Calculate the maximum bending stress. Take $E = 200 \frac{GN}{m^2}$

(9 Marks)



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2016/2017

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE IN ELECTRICAL AND ELECTRONIC ENGINEERING AND BACHELOR
OF SCIENCE IN BIOMEDICAL ENGINEERING

EEE 212: MECHANICS OF MACHINES

DATE: Thursday 15th December 2016

TIME: 11.00a.m-1.00p.m

INSTRUCTIONS:

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Question 1 (30 Marks)

a) Define the following terms: (8 Marks (1 each))

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- Dynamics
- Kinetics
- Kinematics



b) Define a force and state the characteristics required to completely define the force. (5 Marks)

c) A steel rod of 3 cm diameter is enclosed centrally in a hollow copper tube of external diameter 5 cm and internal diameter of 4 cm. The composite bar is then subjected to an axial pull of 45kN. If the length of each bar is

equal to 15 cm, determine: (E of steel = $2.1 \times 10^5 \text{ N/mm}^2$ and for copper = $1.1 \times 10^5 \text{ N/mm}^2$)

- i. Stresses in the rod and the tube
- ii. Load carried by each bar - (7 Marks)

d) A vehicle to be used for establishing a space workshop consists of two parts A and B that are to be separated during flight. At the instant prior to separation, the combined vehicle has a velocity

$$\hat{v} = 5000\hat{i} + 4000\hat{j} + 3000\hat{k} \text{ m/s}$$

At time t after separation, measurements taken on the separated portion A indicated its acceleration components in the three rectangular coordinate directions to be; $a_x = 4.8t \text{ m/s}^2$, $a_y = 1.2t^2 \text{ m/s}^2$, $a_z = 2t \text{ m/s}^2$. It is assumed that the velocity of the main part B of the vehicle remains reasonably constant throughout. Determine:

- i. The velocity of the separated portion A; 20 seconds after separation.
 - ii. The distance between the two parts after this time has elapsed.
- (10 Marks)

$$m \cdot a = \int x \cdot a$$

Question 2 (20 Marks)

With the aid sketch of the stress strain graph, explain the behaviour of a mild steel rod in an extensometer. (10 Marks)

- Define shear force and bending moment. (4 Marks)

Define 'factor of safety' and explain its importance. (6 Marks)

Question 3 (20 Marks)

- a) i) State Newton's second law of motion. (1 Mark)
- ii) State D'Alembert's principle. (1 Mark)

b) Two blocks of weight W_1 and W_2 are connected by inextensible wire passing over a smooth pulley as shown in Figure 3(a). If W_1 is greater

than W_2 , find the tension in the string and determine the acceleration of the system. (6 Marks)

c) The two crates in Fig. 3(b) are released from rest. Their masses are $M_A=40\text{kg}$ and $M_B=30\text{kg}$. If the static and kinetic coefficients of friction between crate A and the inclined surface are $\mu_s=0.2$ and $\mu_k=0.15$ respectively, determine:

- Whether the crates will start moving.
- The acceleration of the crates if they are in motion. (12 Marks)

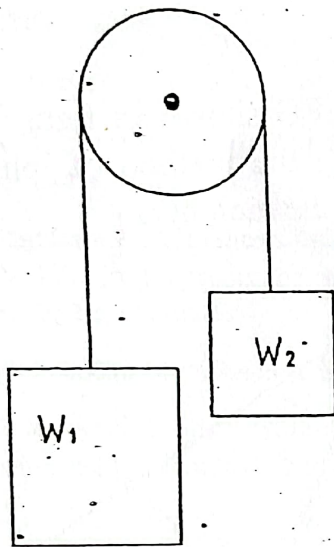


Figure 3 (a)

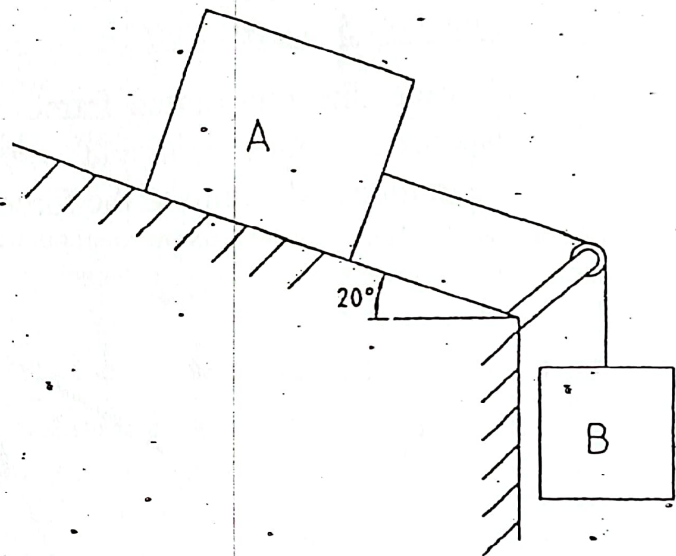


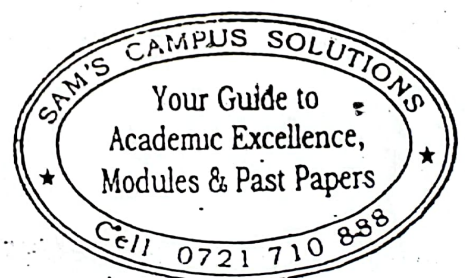
Figure 3 (b)

Question 4 (20-Marks)

a) Figure 4(a) shows the plane area of a sheet metal plate with a circular hole. Determine the coordinates $\bar{x}$ and $\bar{y}$ of the centre of the hole so that this point will be the centroid of the remaining shaded area. (9 Marks)

b) Determine the second moment of area of the cross-section shown in figure 4(b):

- about the horizontal centroid axis.
- about the vertical centroid axis. (11 Marks)



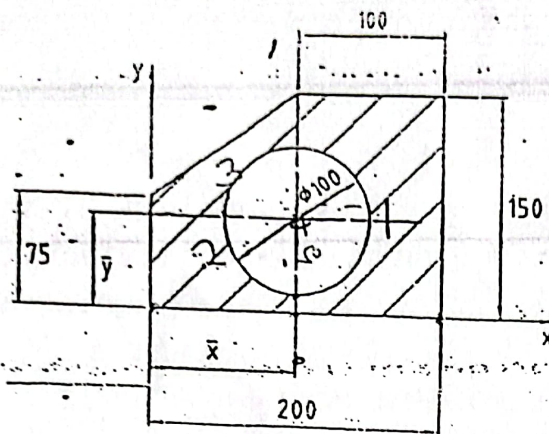


Figure 4 (a)

$$\frac{4\pi r^2}{3}$$

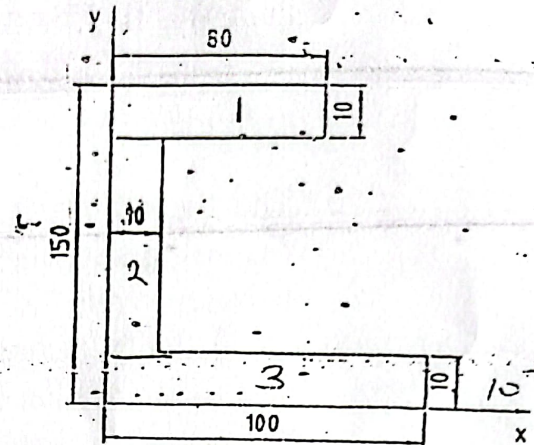


Figure 4 (b)

$$\frac{4\pi r^2}{3}$$

Question 5 (20 Marks)

Consider the pin-jointed frame ABCDEFG shown in figure 5. Determine the support reactions at A and E and using the method of joints, determine the magnitude and nature of the forces in all the members.

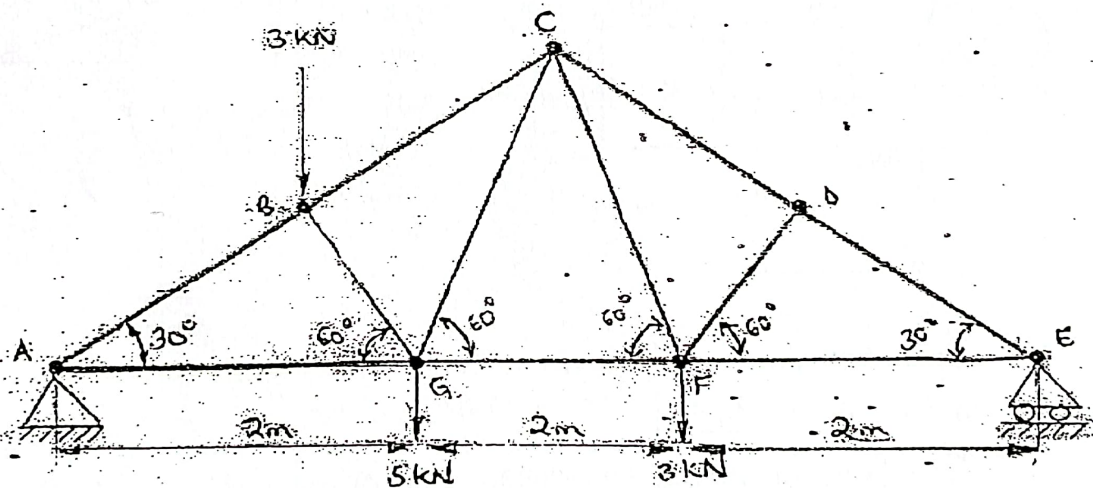
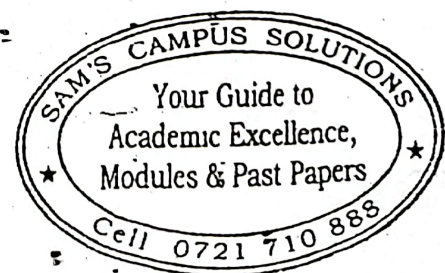


Figure 5



Date: October 2016

Time: 2 hours

1. Define the following terms: (6 Marks)

- Machine
- Particle
- Rigid body
- Kinetics
- Kinematics
- Statics

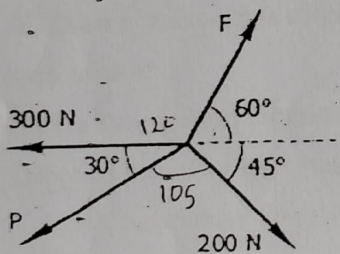


2. Determine the magnitude of P and F in Fig. 1 (a) necessary to keep the force system in equilibrium. (4 Marks)

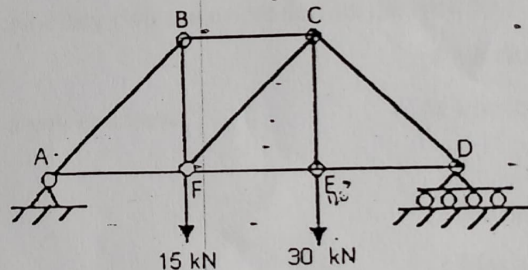
3. The diagram in Fig 1 (b) shows a plane pin-jointed framework subjected to vertical loads of 15kN and 30kN. Solve for the forces and reactions in the frame. The lengths of the vertical and horizontal members are 3m. (14 Marks)

4. Given the loading of the beam as shown in Fig 1 (c), determine the support reactions. (6 Marks)

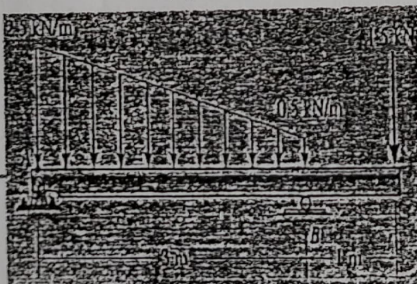
5. Calculate the values and draw the diagrams for shear force and bending moments for the overhanging beam shown in Fig 1 (d). (10 Marks)



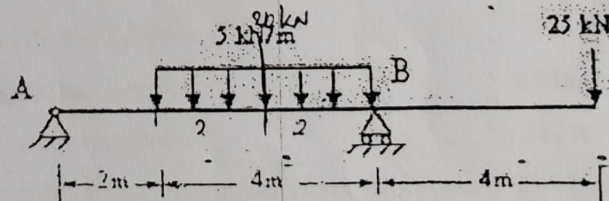
(a) Question 2



(b) Question 3

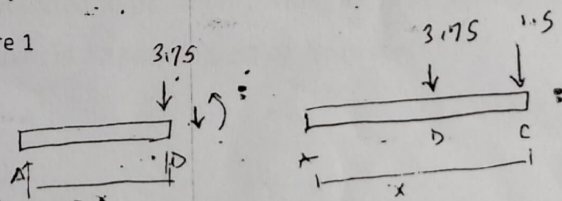


(c) Question 4



(d) Question 5

Figure 1



$$A + 3.75 = 0$$

$$A = -3.75$$

$$A \times 4 = 0$$

$$A + 2.75 + 1.5 = 0$$

$$A = -5.25$$

$$A \times 4 + 3.75 \times 2 + 1.5 \times 4 = 0$$



WATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2019/2020

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN
ELECTRICAL AND ELECTRONIC ENGINEERING AND BACHELOR OF SCIENCE IN
BIOMEDICAL ENGINEERING

EEE 212: MECHANICS OF MACHINES

DATE: Tuesday 4th February 2020

TIME: 11.00a.m -1.00p.m

INSTRUCTIONS

This paper contains FIVE questions

Question ONE is compulsory and carries 30 Marks.

Questions TWO – FIVE carries 20 Marks each. Answer any Two

QUESTIONS ONE (30 MARKS)

- a) (i) Describe how a tensile test is carried out (4 Marks)
- (ii) Draw a stress-strain diagram and describe any two major properties derived from such a curve (4 Marks)
- b) Derive the equation for raising a load on a power screw (4 Marks)
- c) Describe the following types of stresses.
 - i. Compressive Stresses (2 Marks)
 - ii. Shear Stresses (2 Marks)
- d) State the parallel axis theorem (2 Marks)
- e) With the aid of diagrams, differentiate the following
 - i. Point load and uniformly distributed load (2 Marks)
 - ii. A Cantilever and a simply supported beam (2 Marks)
- d) Use diagrams and equations to describe an underdamped vibration system (3 Mark)
- f) A 50kg crate rests on a horizontal plane for which the kinetic co-efficient of friction is 0.3. If the crate does not tip over when subjected to the 400N towing force as shown in Fig. 1f, determine the velocity of the crate in 5 seconds starting from rest (5 Marks)

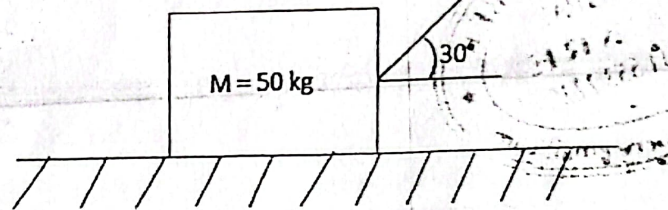


Fig. 1f

QUESTIONS TWO (20 MARKS)

- a) State Newton's second law of motion (2 Marks)
- b) A block of mass 10 kg is attached to a spring and a dashpot as shown in Figure Q2b. The spring constant $k = 1000 \text{ N/m}$ and a damping rate $c = 50 \text{ N} \cdot \text{s/m}$. When the block is at a distance d_0 from the left wall the spring is relaxed. The block is pulled to the right by 0.5m and released.

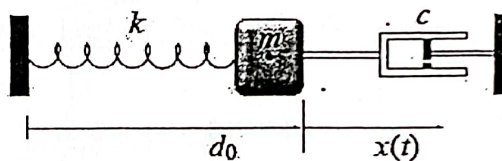


Figure Q2b

- Assuming no initial velocity, find
- The equation of motion of the block (2 Marks)
 - The position of the block at $t = 2 \text{ s}$ (8 Marks)
- g) A steel bar of rectangular cross-section, 10 cm deep and 5cm wide, is bent in the planes of the longer sides. Estimate the greatest allowable bending moment if the bending stresses are not to exceed 150 MN/m^2 in tension and compression

(8 Marks)

QUESTIONS THREE (20 MARKS)

- a) Derive the equation for ratio of belt tensions in a flat belt and pulley system (5 Marks)
- b) The beam shown in Fig. 3b is simply supported at its ends. It carries a uniformly distributed load of 1.5 kN/m over its entire length and two concentrated loads of 2 kN and 5 kN at two points as shown in the figure.

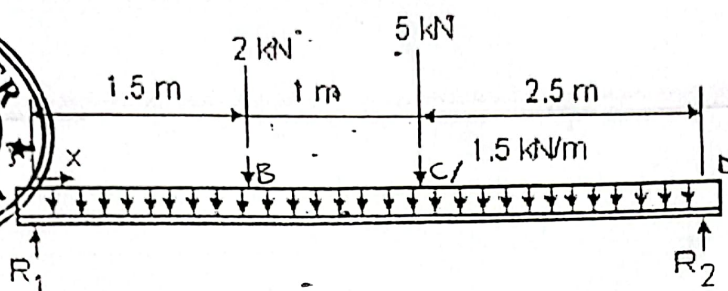


Fig. 3b

- (i) Calculate the values of the reactions R_1 and R_2 (3 Marks)
- (ii) Derive the expressions for the shear force and the bending moment for each segment of the beam. (6 Marks)
- (iii) Sketch the shear force and bending moment diagrams on the graph paper provided, to a linear scale of 1 cm = 1m and a force scale of 1 cm = 10 kN. Neglect the weight of the beam. (6 Marks)

QUESTIONS FOUR (20 MARKS)

- a) Differentiate between kinetics and kinematics (2 Marks)
- b) A steel shaft 35 mm in diameter and 1.2 m long held rigidly at one end has a hand wheel 500 mm in diameter keyed to the other end. The modulus of rigidity of steel is 80 GPa.
 - i. What load applied to tangent to the rim of the wheel produce a torsional shear of 60 MPa? (4 Marks)
 - ii. How many degrees will the wheel turn when this load is applied? (4 Marks)
- c) A single plate clutch with both sides of the plate effective is required to transmit 25 KW at 1600 rpm. The outer diameter of the plate is limited to 300 mm and the intensity of pressure between the plates is not to exceed 0.07 N/mm². Assuming uniform pressure and coefficient of friction as 0.3.
 - (i) Find the inner diameter of the plate and the axial force necessary to engage the clutch. (7 Marks)
 - (ii) Assuming the same dimensions and same total axial thrust, determine the maximum torque that can be transmitted when uniform wear conditions are assumed. (3 Marks)

QUESTIONS FIVE (20 MARKS)

a) Use diagrams to differentiate between cross-sections of structural members with one and two axes of symmetry (2 Marks)

b) (i) Distinguish between a single gear train and a compound gear train (2 Marks)

(ii) The gearing of a machine tool is shown in Fig. Q5b(ii). The motor shaft is connected to gear A and rotates at 975 rpm. The gear wheels B, C, D and E are fixed to parallel shafts rotating together. The final gear F is fixed on the output shaft. What is the speed of gear F? (6 marks)

The numbers of teeth on each gear are as given below:

Gear	A	B	C	D	E	F
No. of teeth	20	50	25	75	26	65

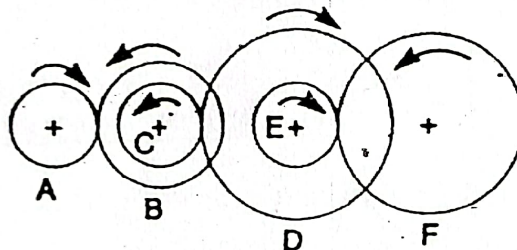


Fig 5b(ii)

- c) The crank of a slider crank mechanism rotates clockwise at a constant speed of 300 rpm. The crank is 150 mm and the connecting rod is 600 mm long. Determine:
- Linear velocity and acceleration of the midpoint of the connecting rod, and
 - Angular velocity and angular acceleration of the connecting rod, at a crank angle of 45° from inner dead centre position. (10 Marks)





KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN ELECTRICAL
AND ELECTRONIC ENGINEERING AND BACHELOR OF SCIENCE IN
BIOMEDICAL ENGINEERING

EEE 212: MECHANICS OF MACHINES

1ST SEMESTER 2020/2021

TIME: 2 Hours

INSTRUCTIONS:

1. You should have the following for this examination.
 - Answer booklet provided
 - Scientific Calculator
 - Drawing tools
2. This paper contains **Five** questions.
3. Attempt a total of **Three** questions.
4. Question One is **COMPULSORY** and carries **30 marks**
5. Be clear in expressions and neat in your working.
6. Begin each question on a fresh page.

Question One (Compulsory) 30 Marks

a) Define the following terms

- i. stress
- ii. strain
- iii. shear modulus (3 marks)

b) ~~For~~ Show that for a prismatic bar loaded axially in tension by a force P the deformation δ is

given by $\delta = \frac{PL}{AE}$ (5 marks)

c) With the aid of sketches, illustrate the following types of beam loadings.

i. cantilever beam

ii. simply supported beam

(4 marks)

d) Differentiate between kinetics and kinematics

(3 marks)

e) Explain the degrees of freedom for a body in space.

(3 marks)

f) A triangular linkage is regarded as stable. Explain stating any four practical applications in engineering.

(3 marks)

g) List the **three** types of free vibrations.

(3 marks)

h) With the aid of sketches, highlight the types of belts used in power transmission

(6 marks)

Question Two

a) Illustrate any four beam cross-sections

(2 marks)

b) Show that the moment of inertia of a rectangular section of width b and depth d is given by

$$\frac{bd^3}{12}$$

(6 marks)

c) Figure Q2 shows a section cross-section. Locate the COG

(2 marks)

d) Describe any **three** motions a body may possess.

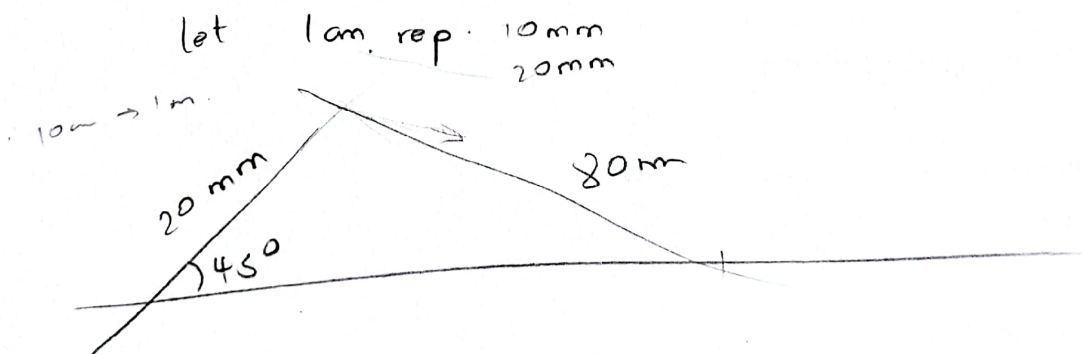
(3 marks)

e) The crank of a steam engine makes an angle of 45° with the horizontal and rotates at 100rpm. The lengths of crank and connecting rod are 20mm long and 80mm respectively. Graphically locate the instantaneous centre for the connecting rod at the given crank position and determine;

i. the velocity of the piston

ii. angular velocity of connecting rod and the crank.

(7 marks)



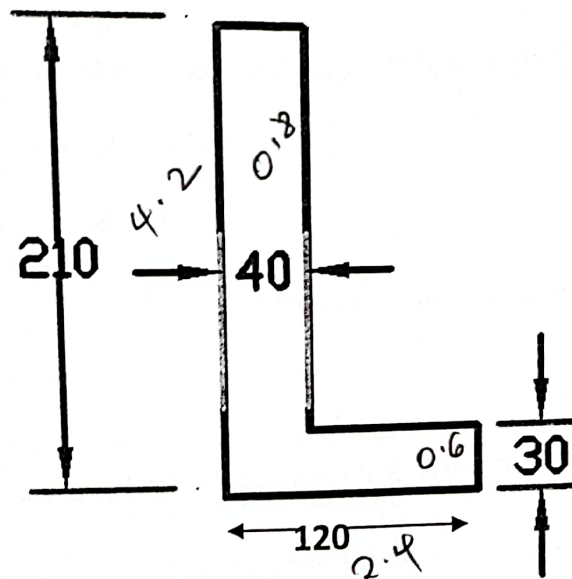


Figure Q2

Question Three

a) Compare gear drive and belt drive systems in power transmission. (4 marks)

b) With the aid of sketches, explain the choice of gears for connecting the following;

- shafts having perpendicular axis
- shafts whose axis are collinear.
- transmission of linear motion using gears

(6 marks)

c) Figure Q3 shows a simply supported beam subjected to an inclined load of 15kN as shown. Draw axial force, shear force and bending moment diagrams for the system. State the greatest value of shear force and bending moment. (10 marks)

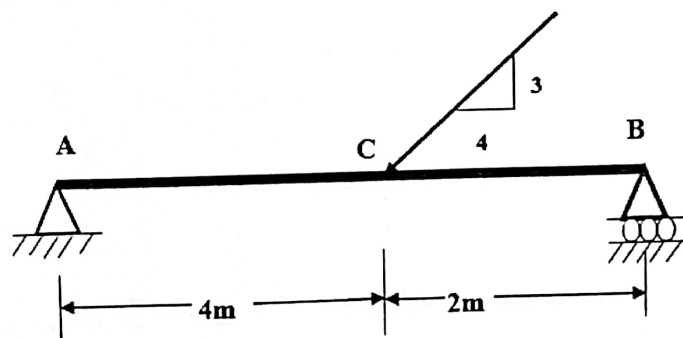


Figure Q3

Question Four

a) (i) Show that for pure bending of a straight uniform rectangular beam having breadth b and depth h

$$\frac{M}{I} = \frac{E}{R} = \frac{\sigma}{y}$$

and define the terms.

(8 marks)

(ii) State any four assumptions made in (i) above.

(2 marks)

b) i) Describe slip as used in belt drives and its effect on velocity ratio. (4 marks)

ii) A double reduction gearing system is shown in **Figure Q4**. The power is transmitted from a motor shaft to output shaft. The motor shaft is connected to gear A rotates in the clockwise direction at 1500 r.p.m. The output shaft is connected to gear D. Gears B and C are mounted on the same shaft. The number of teeth on each of the gear A, B, C and D are 40, 50, 30 and 60 respectively. For the system, determine;

- velocity ratio
- the speed and direction of rotation of gear D
- the driving gears or drivers

(6 marks)

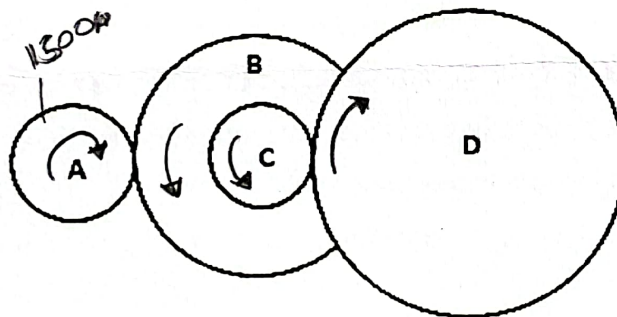


Figure Q4

Question Five

a) Using **Figure Q2**, determine the moment of inertia about axis passing through its CG and parallel to the base. (6 marks)

b) A rectangular cross-section beam has a depth of 200mm and width 24mm and is subject to a bending moment of 2.5kN m. Taking $E = 206 \text{ GN/m}^2$, Determine the maximum stress in the beam (4 marks)

c) List **any four** power transmission elements

(2 marks)

d) Make sketches of the following

i. open and crossed belt drives

ii. simple and epicyclic gear trains

(6 marks)

e) Explain **two** roles of idler in gear transmission

(2 marks)